

ISSAM SAYYAF

Sensor Systems & Algorithms Engineer · MEMS · Sensor Fusion · Signal Processing · Deep Learning · Embedded C / Python



📍 Nantes, France · Relocating to Grenoble 📞 +33 758 120 733 ✉ issamsayyaf97@gmail.com 🔗 linkedin.com/in/issamsayyaf
🐙 github.com/IssamSayyaf 🌐 issam-sayyaf.com

PROFILE

Sensor systems and algorithms engineer with hands-on experience across the full sensor stack — MEMS inertial (IMU: accelerometer, gyroscope, magnetometer), acoustic / ultrasound (piezoelectric AE, BLE Channel Sounding time-of-flight), GNSS/RF and pressure sensors. PhD work on **Anomaly Detection for Time Series Sensor Data** using deep learning (autoencoders, adversarial generative models) to build robust features from noisy sensor data and interpret algorithm performance. Proven delivery of embedded C and C++ on ARM (STM32, NXP i.MX, Nordic nRF), Python pipelines for data collection and characterization, and instrumented test benches (NI / LabVIEW / SCPI). 10+ IEEE publications. Autonomous, collaborative and energized by cross-functional ML / algorithm / HW / validation work.

PROFESSIONAL EXPERIENCE

Research Engineer — Sensor Fusion, Signal Processing & Machine Learning Nov 2023 – Oct 2026
GEOLoc Lab, Université Gustave Eiffel, France (PhD)

- ▶ Designed anomaly-detection algorithms on multi-sensor platforms (IMU accelerometer + gyroscope, GNSS receivers, 5G modules): Kalman-family loop observers (DLL/PLL/FLL), spectral analysis and classical detection theory for robust detection/classification under white and colored noise
- ▶ Built advanced deep-learning pipelines in Python (PyTorch) — autoencoders, adversarial generative models, isolation forest — for time-series anomaly detection on sensor streams; defined data-collection protocols and engineered features so the information content of sensor data maps to interpretable performance indicators
- ▶ Implemented real-time C and C++ signal-processing chains on ARM (FFT/PSD, filtering, spectral feature extraction); optimized algorithms for embedded deployment and cross-validated on experimental datasets
- ▶ Authored 10+ IEEE publications (Sensors Journal, I2MTC, IPIN, MetroLivEnv); mentored interns on sensor characterization, measurement chains and ML workflows

Firmware & Algorithms Engineer — Ultrasound/ RF Ranging (BLE Channel Sounding) Apr 2026 – Oct 2026
NAV4YOU, Nantes, France (part-time)

- ▶ Implemented real-time time-of-flight / ranging algorithms (IFFT, MUSIC, ESPRIT super-resolution) on Nordic nRF54L15 under Zephyr RTOS for BLE 6.0 Channel Sounding indoor positioning
- ▶ Developed hardware configuration routines, antenna driving and synchronization of ranging acquisitions; production-grade C with Git and code-coverage instrumentation
- ▶ Characterized, debugged and evaluated algorithm performance on bench; compared experimental results against reference measurements and documented error budgets

EDUCATION

NOV 2023 – JUN 2026
PhD, Signal Processing & Embedded Systems
Université Gustave Eiffel, France
Advanced deep learning for time-series sensor data (IMU, GNSS, 5G). Estimation theory & modern DL.

SEP 2021 – OCT 2023
MSc, IoT & Telecommunications
Università della Calabria, Italy
110/110 cum laude · Acoustic/ultrasound sensing, signal processing, instrumentation

SEP 2014 – SEP 2019
BEng, Electronics & Communications
University of Aleppo, Syria
88.56% with distinction · Electronics, control theory, signal processing

TECHNICAL SKILLS

Sensors & MEMS

Multi-modal sensor integration & characterization

IMU (accel + gyro) Magnetometer
Microphone / Acoustic
Ultrasound / Time-of-Flight Piezoelectric (AE)
Pressure GNSS / RF ADC high-resolution
SPI / I2C / UART

Signal Processing & Algorithms

Estimation, detection, classification

Kalman-family observers DLL / PLL / FLL
FFT / PSD / filtering MUSIC / ESPRIT / IFFT
Feature engineering Detection / Classification
Algo optimization

Machine Learning & Deep Learning

Advanced DL for time-series sensor data

PyTorch scikit-learn Autoencoders
Adversarial generative models (GAN)
Isolation Forest Time-series modeling
TinyML / Edge AI

Embedded SW

Industrial-grade embedded development

Embedded Systems Engineer — MEMS & Industrial Sensor Integration

Jun 2023 – Oct 2025

Hexabitz, USA (remote) — Industrial IoT Products

- ▶ Wrote 10+ production Linux drivers (IIO, serdev, platform) in C for MEMS / industrial sensors — IMU (accel + gyro), audio codec, GNSS receiver, pressure-capable peripherals, 4G modem — over SPI, I2C, UART and ADC on ARM platforms (STM32MP1, NXP i.MX93)
- ▶ Built and customized full embedded Linux systems with Yocto / OpenEmbedded: meta-layer development, BSP integration, kernel configuration and customization, device-tree bring-up and rootfs tailoring for product images
- ▶ Designed low-level acquisition interfaces: register-level configuration, sensor calibration, time synchronization and multi-sensor streaming integrated into the Linux software stack
- ▶ Followed industrial SW practices: Git workflows, code reviews, CI/CD (GitHub Actions), Docker cross-compilation and documentation for internal teams, partners and suppliers

Research Engineer — Acoustic/Ultrasound Sensing, Mechanical Integration & LabVIEW

Oct 2022 – Oct 2023

Università della Calabria, Italy (MSc research, 110/110 cum laude)

- ▶ Modeled and experimentally characterized ultrasonic elastic-wave propagation through mechanical structures using piezoelectric acoustic-emission (AE) sensors — vibration propagation, filter response, acoustic coupling and analog signal conditioning — for structural health monitoring
- ▶ Validated best mechanical integration of AE sensors on real samples: NI DAQ + signal generator driven in LabVIEW, oscilloscope and spectrum analyzer controlled via SCPI, with cross-validation between acquisition paths
- ▶ Applied signal-processing and classification techniques (FFT, band-pass filtering, transient detection, feature extraction) in LabVIEW and Python to discriminate crack-emission events from structural/ambient noise
- ▶ Transposed the full NI bench to a low-cost autonomous edge node (STM32 + FreeRTOS + MQTT) to demonstrate product-grade embedded integration. Published at IEEE MetroLivEnv 2023

Teaching Assistant — Signal Processing & Electronics

Sep 2020 – Aug 2021

University of Aleppo, Syria

- ▶ Led 50+ lab sessions (150+ students) in signal processing, electronics and embedded systems; trained students on lab instrumentation and sensor measurement methodology

KEY PROJECTS

Advanced Deep Learning for Time-Series Sensor Data — IMU / GNSS / 5G

2023 – 2026

PyTorch pipeline combining classical sensor-domain features (Kalman-family loop observables, spectral descriptors) with advanced deep-learning models (autoencoders, adversarial generative models) to detect and classify anomalies on multi-modal sensor streams — published in IEEE Sensors Journal (2025).

Ultrasound/Acoustic SHM Bench — Piezoelectric AE + LabVIEW + NI + SCPI

2022 – 2023

End-to-end acoustic sensing platform: mechanical integration of piezoelectric AE sensors on structural samples, NI DAQ + signal generator in LabVIEW, oscilloscope and spectrum analyzer in SCPI, analog front-end, vibration/acoustic-coupling characterization and FFT-based crack detection. Transposed to an STM32 + FreeRTOS + MQTT edge node. IEEE MetroLivEnv 2023.

C

C++

Yocto / OpenEmbedded

Linux kernel & drivers (IIO, serdev)

STM32 / ARM Cortex-A/M

Nordic nRF54L15

FreeRTOS / Zephyr

GDB / JTAG

Python & SW Tooling

Data pipelines, team practices

Python (numpy, scipy, pandas)

Git / GitLab / GitHub

CI/CD (GitHub Actions)

Docker

Code coverage

Instrumentation & Validation

Bench design, characterization, debug

LabVIEW

NI DAQ

Oscilloscope

Spectrum analyzer

SCPI / GPIB

Vibration / acoustic coupling

CERTIFICATIONS

Deep Learning & Edge AI

- ✓ Deep Learning Specialization (DeepLearning.AI)
- ✓ TinyML (Harvard / edX)

Bootlin (2024) — Embedded Linux

- ✓ Linux Kernel Driver Development
- ✓ Yocto Project Development
- ✓ Embedded Linux Development

LANGUAGES

English	C1 – Fluent (professional)
French	B1/B2 – Intermediate-advanced
Arabic	Native

ADDITIONAL STRENGTHS

10+ IEEE publications · 5+ international conferences · Intern supervision · 150+ students trained · Available September 2026 (post-defense) · Relocating to Grenoble

Custom Yocto BSP — STM32MP157F Board Bring-up

2023 – 2025

Full custom board bring-up: device tree configuration, Yocto meta-layer creation, kernel driver integration (IIO, serdev), secure boot chain (TF-A/OP-TEE/U-Boot), SWUpdate OTA with LUKS. Automated image build via CI/CD pipeline.

NXP i.MX93 Yocto Platform — Industrial IoT Gateway

2024 – 2025

BSP development for NXP i.MX93 (Cortex-A55): Yocto layer structuring, kernel driver development for multi-sensor acquisition, systemd service management, and Docker-based cross-compilation environment.

SELECTED PUBLICATIONS (10+ TOTAL)

- **M. I. Sayyaf** et al., "Time-Series Anomaly Detection for Sensor Data: Models, Metrics, and Methodologies," *IEEE Sensors Journal*, vol. 25, 2025
- +9 IEEE publications — see scholar.google.com · Conferences: IEEE I2MTC 2026, IPIN 2026, MetroLivEnv 2023